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Q1. What are LCD and TFT technologies? How do these two differ?

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A1. An LCD (short for 'liquid crystal display') is a thin, flat display device made up of any number of colour or monochrome pixels arrayed in front of a light source or reflector.

In colour TVs based on LCD technology, the coloured image is produced by selectively filtering a white light provided by a series of cold-cathode fluorescent lamps at the back of the screen. Some displays use white or coloured LEDs instead.

Millions of individual LCD shutters, arranged in a grid, open and close to allow a metered amount of the white light through. Each shutter is paired with a coloured filter to remove all but the red, green or blue (RGB) portion of the light from the original white source. Each shutter-filter pair forms a single sub-pixel. The sub-pixels are so small that when the display is viewed from even a short distance, the individual colours blend together to produce a single spot of colour—a pixel. The shade of colour is controlled by changing the relative intensity of the light passing through the sub-pixels.

LCD TVs are thinner and lighter than CRT TVs of similar display size, and available in much larger sizes as well. Screens are available in sizes ranging from 36 cm (14 inches) to 165 cm (65 inches). Experimental and limited-run sets are available in sizes of over 254 cm (100 inches). This combination of features has made LCDs more practical than CRTs for many roles. Very little power consumption makes these suitable for use in battery-powered systems. Picture quality is comparable to CRT TVs, but picture sharpness and geometry are much better. No doubt, LCD TVs are expensive—roughly three times the cost of a CRT TV—but for many applications where space, weight and eye fatigue are the issues for consideration, ben-

efits over-ride the cost factor.

LCDs have their disadvantages as well. These are relatively inefficient in terms of power use per display size. Modern LCD sets have attempted to address the power use issue through a process known as 'dynamic lighting' in place of the existing backlighting system. The major benefit of dynamic lighting is the improvement in contrast ratio and shadow detail.

Also, LCDs have lower refresh rates and slower response times. Although improvements have been made in the latest displays, yet the system is not really fast enough because the pixel will still be switching while the frame is being displayed. Conventional CRT and plasma displays have much faster response times. Several newer technologies like OLED, FED and SED are also vying to enter the large-screen television market by overcoming the drawbacks of LCDs.

TFT-LCD is a variant of LCD that uses thin-film transistor (TFT) technology. In general, a display is composed of a grid (matrix) of picture elements called 'pixels.' Millions of these pixels together create an image on the display. TFT-based displays use a transistor for each pixel on the screen. The transistors (TFTs) act as switches to individually turn each pixel 'on' (light) or 'off' (dark). TFTs are active elements, arranged in a matrix, on the display, hence the name 'active-matrix TFT' (AM TFT). Rows and columns of the grid are managed electronic driver circuits that push signals onto the panel's row and column addressing lines. Two basic classes of the device are used—column drivers and row or gate drivers. Electrical current that illuminates the display is turned on and off at a faster rate, which makes the display brighter and shows the motion as smooth.

The technique employed for colour generation is complex. It is done by tripling the number of TFT/capacitor/electrode cells (pixels) to produce a three-colour RGB optical filter layer. In other words, each pixel is made up of